

Work these problems to prepare for Quiz 5 (Sections 14.6–14.7). Show your work. Solutions are on the back.

1. Let $f(x, y) = x^2y - 3y^2$.

- (a) Find the directional derivative of f at the point $(1, 2)$ in the direction toward the point $(4, 6)$.
(b) Find the maximum rate of change of f at $(1, 2)$ and the unit direction in which it occurs.

Solution:

$\nabla f = \langle 2xy, x^2 - 6y \rangle$, so $\nabla f(1, 2) = \langle 4, -11 \rangle$.

(a) Direction toward $(4, 6)$: $\mathbf{v} = \langle 3, 4 \rangle$, so $\hat{u} = \frac{1}{5}\langle 3, 4 \rangle$.

$$D_{\hat{u}}f(1, 2) = \langle 4, -11 \rangle \cdot \frac{1}{5}\langle 3, 4 \rangle = \frac{12 - 44}{5} = \boxed{-\frac{32}{5}}$$

(b) Maximum rate of change $= |\nabla f(1, 2)| = \sqrt{16 + 121} = \boxed{\sqrt{137}}$.

It occurs in the direction $\hat{u} = \frac{1}{\sqrt{137}}\langle 4, -11 \rangle$.

2. Find the equation of the tangent plane to the surface $x^2 + y^2 - z^2 = 4$ at the point $(2, 1, -1)$.
Hint: Write $F(x, y, z) = x^2 + y^2 - z^2$ and use ∇F .

Solution:

Let $F(x, y, z) = x^2 + y^2 - z^2$. Then:

$$\nabla F = \langle 2x, 2y, -2z \rangle$$

At $(2, 1, -1)$:

$$\nabla F(2, 1, -1) = \langle 4, 2, 2 \rangle$$

Tangent plane:

$$4(x - 2) + 2(y - 1) + 2(z + 1) = 0$$

$$\boxed{4x + 2y + 2z = 8} \quad \text{or equivalently } 2x + y + z = 4$$

3. Find all critical points of $f(x, y) = 3x^2 + x^2y + y^3 + 6y^2$ and classify each as a local maximum, local minimum, or saddle point.

Solution:

$$f_x = 6x + 2xy = 2x(3 + y) = 0 \quad f_y = x^2 + 3y^2 + 12y = 0$$

From $f_x = 0$: either $x = 0$ or $y = -3$.

Case 1: $x = 0$. Then $f_y = 3y^2 + 12y = 3y(y + 4) = 0$, so $y = 0$ or $y = -4$.

Critical points: $(0, 0)$ and $(0, -4)$.

Case 2: $y = -3$. Then $f_y = x^2 + 27 - 36 = x^2 - 9 = 0$, so $x = \pm 3$.

Critical points: $(3, -3)$ and $(-3, -3)$.

Second derivatives: $f_{xx} = 6 + 2y$, $f_{yy} = 6y + 12$, $f_{xy} = 2x$.

$$D = f_{xx}f_{yy} - (f_{xy})^2.$$

Point	f_{xx}	D	Classification
$(0, 0)$	6	$(6)(12) - 0 = 72 > 0$	local min
$(0, -4)$	-2	$(-2)(-12) - 0 = 24 > 0$	local max
$(3, -3)$	0	$(0)(-6) - 36 = -36 < 0$	saddle
$(-3, -3)$	0	$(0)(-6) - 36 = -36 < 0$	saddle

4. Find the absolute maximum and absolute minimum values of $f(x, y) = x + y - xy$ on the triangular region D with vertices $(0, 0)$, $(6, 0)$, and $(0, 6)$.

Solution:

Interior: $f_x = 1 - y = 0 \Rightarrow y = 1$. $f_y = 1 - x = 0 \Rightarrow x = 1$.

Critical point: $(1, 1)$. Check: $1 + 1 < 6$ so it is inside D . $f(1, 1) = 1 + 1 - 1 = 1$.

Edge 1: $y = 0$, $0 \leq x \leq 6$. $f(x, 0) = x$. Min = 0 at $(0, 0)$; max = 6 at $(6, 0)$.

Edge 2: $x = 0$, $0 \leq y \leq 6$. $f(0, y) = y$. Min = 0 at $(0, 0)$; max = 6 at $(0, 6)$.

Edge 3: $x + y = 6$, i.e., $y = 6 - x$, $0 \leq x \leq 6$.

$$f(x, 6 - x) = x + (6 - x) - x(6 - x) = 6 - 6x + x^2$$

$$f'(x) = -6 + 2x = 0 \Rightarrow x = 3, \quad y = 3$$

$$f(3, 3) = 3 + 3 - 9 = -3.$$

Comparing all values: $f(0, 0) = 0$, $f(6, 0) = 6$, $f(0, 6) = 6$, $f(1, 1) = 1$, $f(3, 3) = -3$.

Absolute max = 6 at $(6, 0)$ and $(0, 6)$. Absolute min = -3 at $(3, 3)$.

5. Find the point on the plane $3x + 2y + z = 14$ that is closest to the origin.

Hint: Minimize $D^2 = x^2 + y^2 + z^2$ after eliminating z .

Solution:

From the plane: $z = 14 - 3x - 2y$. Substitute into D^2 :

$$f(x, y) = x^2 + y^2 + (14 - 3x - 2y)^2$$

$$f_x = 2x + 2(14 - 3x - 2y)(-3) = 2x - 6(14 - 3x - 2y) = 20x + 12y - 84$$

$$f_y = 2y + 2(14 - 3x - 2y)(-2) = 2y - 4(14 - 3x - 2y) = 12x + 10y - 56$$

Setting both equal to zero:

$$20x + 12y = 84 \quad \Rightarrow \quad 5x + 3y = 21$$

$$12x + 10y = 56 \quad \Rightarrow \quad 6x + 5y = 28$$

From the first: $y = \frac{21 - 5x}{3}$. Substitute into the second:

$$6x + 5 \cdot \frac{21 - 5x}{3} = 28 \quad \Rightarrow \quad 18x + 105 - 25x = 84 \quad \Rightarrow \quad x = 3$$

Then $y = \frac{21 - 15}{3} = 2$ and $z = 14 - 9 - 4 = 1$.

Closest point: $(3, 2, 1)$
